

An Auto-Ranging Capacitance Meter on the ATmega328P

Charge-time measurement, supply-independent dual-threshold timing, and Bluetooth telemetry

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A digital capacitance meter is built on an ATmega328P. An unknown capacitor is charged through a known resistor and the time taken to cross two supply-referenced thresholds is timed in hardware; choosing the thresholds at one-third and two-thirds of the rail makes the result independent of supply voltage. A 16-channel analog multiplexer gives automatic ranging over seven decades, and a two-point calibration removes the residual systematic error. Measurements from 100 pF to 470 μ F are presented, and the dominant low-range limit — multiplexer on-resistance — is identified and quantified.

1 Introduction

A microcontroller cannot read capacitance directly, but it can **measure time** with great precision. This instrument therefore converts the unknown capacitance into a time interval — using nothing more exotic than a resistor charging a capacitor — and counts that interval with a hardware timer. The design’s value lies in arranging the measurement so the result depends only on quantities we know well (a resistor and a clock) and not on those we do not (the exact supply, the comparator’s switching point, or temperature).

The report is organised as a standard development narrative: the measurement **theory** and the case for automatic ranging (§2); the **design and implementation** — architecture, pin-level operation, and the range-search algorithm (§3); the **calibration methodology** (§4); the measured **results** (§5); a **discussion and error analysis**, including a calibration simulation (§6); a **comparison with professional instruments** (§7); and **conclusions** (§8).

2 Theory: the measurement principle

2.1 The RC step response

Consider a capacitor C charged from a constant source V_{cc} through a series resistor R . The node voltage $v(t)$ obeys the first-order linear differential equation

$$RC \frac{dv}{dt} + v(t) = V_{cc},$$

whose solution, for an initially discharged capacitor, is the well-known exponential approach

$$v(t) = V_{cc} (1 - e^{-t/RC}). \quad (1)$$

The product $\tau = RC$ is the **time constant**: the time for the node to reach $1 - e^{-1} \approx 63.2\%$ of V_{cc} . Because the output is taken *across the capacitor*, the network is a first-order **low-pass filter** — the capacitor voltage cannot change instantaneously, so it integrates the input and yields the smooth curve of Equation 1. (Taking the output across the resistor would give a high-pass response — a decaying spike — which carries no usable timing information for our purpose.)

2.2 Inverting the curve to recover capacitance

Rearranging Equation 1, the time taken to reach a fraction of the supply is

$$t = -RC \ln \left(1 - \frac{v}{V_{cc}} \right). \quad (2)$$

A single-threshold scheme would charge from zero and stop the clock when v crosses one reference. This works, but Equation 2 shows that the result then depends explicitly on V_{cc} through the ratio v/V_{cc} — any drift or noise in the supply propagates directly into the capacitance.

2.3 Removing the supply dependence: the two-threshold method

We instead measure the *interval* between two thresholds, V_{low} and V_{high} , crossed as the node charges. Applying Equation 2 at each threshold and subtracting,

$$\Delta t = t_{\text{high}} - t_{\text{low}} = RC \ln \frac{V_{cc} - V_{\text{low}}}{V_{cc} - V_{\text{high}}}. \quad (3)$$

Now choose the thresholds as fixed fractions of the supply, generated by a resistor divider so that they *track* V_{cc} :

$$V_{\text{low}} = \frac{1}{3} V_{cc}, \quad V_{\text{high}} = \frac{2}{3} V_{cc}.$$

Substituting into Equation 3, the supply cancels completely:

$$\Delta t = RC \ln \frac{V_{cc} - \frac{1}{3} V_{cc}}{V_{cc} - \frac{2}{3} V_{cc}} = RC \ln \frac{2/3}{1/3} = RC \ln 2. \quad (4)$$

The capacitance is therefore

$$C = \frac{\Delta t}{R \ln 2} = \frac{\Delta t}{0.693 R} \quad (5)$$

This is a remarkably clean result. The measured capacitance depends only on the **known resistor**, the **measured time interval**, and a **dimensionless constant** $\ln 2$. The supply voltage, which is the least stable quantity in a hobby circuit, has vanished. These particular thresholds — one-third and two-thirds of the rail — will be familiar: they are the same levels used internally by the classic 555 timer, and for the same reason.

2.4 The need for automatic ranging

Equation 5 suggests that *any* resistor would do. In practice a fixed resistor measures only a narrow band of capacitance, for two independent reasons — one arising from the digital timer, the other from the physics of the RC circuit.

2.4.1 The mathematical argument: a bounded counter

The interval Δt is not measured continuously but counted in discrete timer ticks of duration t_{tick} :

$$N = \frac{\Delta t}{t_{\text{tick}}}, \quad N_{\text{min}} \leq N \leq N_{\text{max}}.$$

Both bounds are real. At the **lower** end, the quantization error is ± 1 tick, i.e. a *relative* error of $\pm 1/N$; a count of $N = 5$ carries a 20% uncertainty, so we require $N \gtrsim 200$ for sub-percent resolution. At the **upper** end, the 16-bit timer saturates at $N_{\text{max}} = 65535$; beyond this the counter overflows and the reading is meaningless.

A single resistor therefore admits only the capacitance band

$$C \in \left[\frac{N_{\text{min}}}{0.693 R} t_{\text{tick}}, \frac{N_{\text{max}}}{0.693 R} t_{\text{tick}} \right],$$

a span of $N_{\text{max}}/N_{\text{min}} \approx 65535/200 \approx 330$, or about **2.5 decades**. To cover the design target of 1 pF–1 μ F (six decades) — let alone the picofarad-to-millifarad range the hardware actually reaches — one resistor is mathematically insufficient. Since $\Delta t \propto RC$, keeping Δt inside the good window as C varies requires scaling R *inversely* with C . Because C is unknown a priori, the instrument must **try resistors until the count lands in range** — this is automatic ranging. A geometric ladder of resistors, each a fixed multiple of the last, tiles the whole span with overlapping windows.

2.4.2 The physical argument: matching the timescale

The same conclusion follows from physics. The time constant $\tau = RC$ sets how quickly the capacitor charges. If τ is *too short* (small RC) the node reaches the thresholds faster than the comparator's propagation delay and the timer's resolution can follow — one would be timing electrical artefacts, not the capacitor. If τ is *too long*, a measurement takes seconds to minutes and the counter overflows.

Every instrument must arrange for the physical event it measures to unfold on a timescale its sensor can resolve. Auto-ranging selects R so that charging occupies a *comfortable* window — long enough to accumulate many counts (good resolution), short enough to finish quickly and not overflow. The principle is identical to a multimeter's range switch (200 mV / 2 V / 20 V ...) or to choosing appropriate units on a stopwatch when timing a sprinter versus a glacier: one matches the scale of the tool to the magnitude of the quantity, so that the full resolution is used without running off the end of the scale.

3 Design and implementation

3.1 System architecture

Cx	R(range)	node	analog comparator	Timer1 input capture
CD4067 mux (auto-range R)			1/3 & 2/3 Vcc divider	
7-seg display	SPI/595		HC-06	USART (Bluetooth telemetry)

Timing front-end. The node drives the on-chip analog comparator's positive input; the negative input is switched between the two reference taps through the ADC multiplexer, so the threshold is changed by a single register write. The comparator output is routed internally to the Timer1 *input-capture* unit, which latches the count in hardware at each crossing, free of interrupt-latency jitter.

Range selection and references. A CD74HC4067 16-channel multiplexer places one of a ladder of resistors between the charge pin and the node, selected by four digital lines. Three equal resistors from V_{cc} to ground form the $\frac{1}{3}V_{cc}$ and $\frac{2}{3}V_{cc}$ taps; since these and the charge source share the rail, the supply cancels (Equation 4) and the divider ratio need not even be precise — calibration absorbs it.

Display and telemetry. A four-digit seven-segment display is multiplexed through 74-series shift registers over hardware SPI, and results are streamed as ASCII over UART to an HC-06 Bluetooth module, which also lets a host trigger measurements remotely for collecting the calibration dataset below.

3.2 Pin assignment and the capture sequence

Two microcontroller pins do the analogue work. **PB0** is the charge/discharge drive: as a *switchable* supply it is driven HIGH ($\approx V_{cc}$) to charge C_x through the range resistor, and LOW to discharge it — which is why the node is not tied to V_{cc} directly (we would lose the ability to reset). **PD6** is the comparator's positive input (AIN0); being high-impedance it *senses* the node voltage without disturbing the charging.

The comparator's negative input is, by default, the pin AIN1 (PD6's neighbour, PD7). We do **not** use it: setting the **ACME** bit (with the ADC disabled) reroutes the negative input to the **ADC multiplexer**, so the threshold is chosen in firmware by writing **ADMUX** — **PC4** for $\frac{1}{3}V_{cc}$ or **PC3** for $\frac{2}{3}V_{cc}$. This frees the AIN1 pin (PD7), which is then reused as a plain output: the 4067 channel-select bit **S3** (S0–S2 being PC0–PC2).

A measurement proceeds in two distinct phases — only one of which is *timed*:

Step	Reference (–)	Node sensed by	Timer
Discharge (reset)	PC4 (Vcc)	poll ACO level until 0	—
Charge → t_low	PC4 (Vcc)	rising-edge capture	latched
Charge → t_high	PC3 (Vcc)	rising-edge capture	latched

The discharge is a *software level poll* — we wait until the comparator reports the node has fallen below $\frac{1}{3}V_{cc}$ — which adapts automatically to any RC and is never timed. Only the charge is timed: with the input-capture edge set to *rising* (**ICES1=1**), the comparator output routed internally to Timer1 (**ACIC**) latches **TCNT1** in hardware as the node crosses each threshold, free of interrupt latency. The interval is $\Delta t = t_{\text{high}} - t_{\text{low}}$, which by Equation 4 equals $RC \ln 2$.

In firmware the interval is a tick count, $\Delta t = N t_{\text{tick}}$ with $t_{\text{tick}} = 1 \mu\text{s}$ (8 MHz, $\div 8$ prescaler), and the reported value uses the calibrated affine form

$$C[\text{pF}] = N \frac{\text{CAL_K}}{R} + \text{CAL_B},$$

where $\text{CAL_K} = 1/p$ folds the clock and $\ln 2$, CAL_B is the offset, and R is the selected channel's resistance (ideally $R_{\text{nom}} + R_{\text{on}}$).

3.3 The range-search algorithm

Selecting the channel needs no knowledge of C . The decision is a single test on the tick count N : a channel is **correct** when $N \in [N_{\text{min}}, N_{\text{max}}] = [200, 58000]$ — above which the count overflows the 16-bit timer, below which ± 1 -tick quantization exceeds tolerance. Since $N \propto RC$ and C is fixed, the instrument simply changes R until N falls in this window.

The first implementation was a **warm-started, direction-guided linear search**: start at the last-used channel (`g_range`, initialised mid-ladder), and step one channel toward smaller R if N is too large or larger R if too small, stopping at the first in-window channel. With only eleven geometrically-spaced channels and a remembered start, repeated similar capacitors resolve in zero or one step. Its weakness is a *cold* jump between very different capacitors, which walks one channel at a time across the ladder.

This is removed by an **interpolation step**. Because $N \propto RC$, a single probe predicts the ideal resistance directly:

$$R_{\text{target}} = R_{\text{ch}} \frac{N_{\text{target}}}{N_{\text{measured}}}, \quad (6)$$

so the search jumps straight to the channel nearest R_{target} and confirms with one short verify step (an overflowed probe is treated as a large N , forcing a downward jump). The result is **hybrid**: the warm-start probe costs nothing for repeated similar caps, and the interpolation jump bounds *any* cold transition to ~ 2 – 3 measurements regardless of how many decades apart the capacitors are — the equivalent of interpolation search over the range ladder.

(Empirical convergence counts for the hybrid search, and the assembled-circuit schematics, will be added once measured on the rebuilt hardware.)

4 Calibration methodology

4.1 The need for, and form of, calibration

Equation 5 is the *ideal* relationship. The realized instrument departs from it through several systematic effects: the internal RC oscillator's frequency (hence t_{tick}) is accurate only to about $\pm 10\%$; the reference ratio is not exactly $\ln 2$; the resistors carry tolerance; the multiplexer adds series resistance; and the comparator and stray capacitance introduce small offsets. Rather than measure each effect separately, we absorb them all by fitting the instrument's response to capacitors of *known* value.

The multiplicative effects (clock, ratio, resistance) scale the slope, while the additive effects (stray capacitance, comparator delay) contribute an offset. The appropriate model is therefore an **affine** relation,

$$C = a \Delta t + b, \quad (7)$$

with slope a (in pF per tick) and intercept b (in pF). This form is not assumed but *derived*: writing the measured interval with both error types present,

$$\Delta t = \underbrace{\frac{\ln 2 \cdot R}{t_{\text{tick}}}}_{\text{slope}} (C + C_{\text{stray}}) + t_{\text{delay}}, \quad (8)$$

and solving for C gives

$$C = \frac{t_{\text{tick}}}{R \ln 2} \Delta t - \underbrace{\left(\frac{t_{\text{delay}} t_{\text{tick}}}{R \ln 2} + C_{\text{stray}} \right)}_{-b},$$

so the **slope** carries the multiplicative errors and the **intercept** lumps the additive ones (C_{stray} and the comparator/mux delay). A purely proportional law ($b = 0$) would force the calibration line through the origin; the measured intercept of $\approx +650$ pF (Fig. Figure 1) is direct evidence that the offset is real and the affine form is the correct one.

4.2 Two-point calibration

Two known capacitors $(\Delta t_1, C_1)$ and $(\Delta t_2, C_2)$ determine the line uniquely:

$$a = \frac{C_2 - C_1}{\Delta t_2 - \Delta t_1}, \quad b = C_1 - a \Delta t_1. \quad (9)$$

A *single* point would fix only the slope and force the line through the origin ($b = 0$); the results of §5 show this is inadequate, because a non-zero offset is present. Two points capture both parameters, so the calibrated line is correct across the whole range rather than at one capacitor only.

4.3 A shortcut from measured resistors

Because each ladder resistor is individually measured, the per-range slope need not be calibrated separately. Writing $a = K/R$ with a single global constant $K = t_{\text{tick}}/\ln 2$, the model becomes

$$C = \frac{\Delta t K}{R} + b, \quad (10)$$

so that calibrating K and b *once* defines every range, each range’s slope following from its measured resistance. This collapses a sixteen-range calibration into the determination of two numbers.

5 Results

5.1 Single-range characterization

The range resistor was first locked to channel 7 ($R = 99.3$ k Ω) and a set of known capacitors measured, each as the trimmed mean of 30 charge cycles (Table 1).

Table 1: Channel-7 characterization. The usable window (3–90 nF) agrees with the predicted span of §2.4.

Capacitor	Δt (ticks)	Within window?
100 pF	12	No — too fast ($\pm 8\%$ quantization)
10 nF (mylar)	733	Yes
“104” ceramic	2802	Yes (≈ 40 nF; DC-bias loss)
50 nF	3579	Yes
100 nF	7788	Yes
1 μ F	overflow	No — too slow

Applying the two-point fit of Equation 9 to the 10 nF and 100 nF points,

$$a = \frac{100000 - 10000}{7788 - 733} = 12.76 \text{ pF/tick}, \quad b = 10000 - 12.76 \times 733 = +650 \text{ pF},$$

equivalently $K = aR \approx 1.267 \times 10^6$ and $b = 650$ pF in Equation 10.

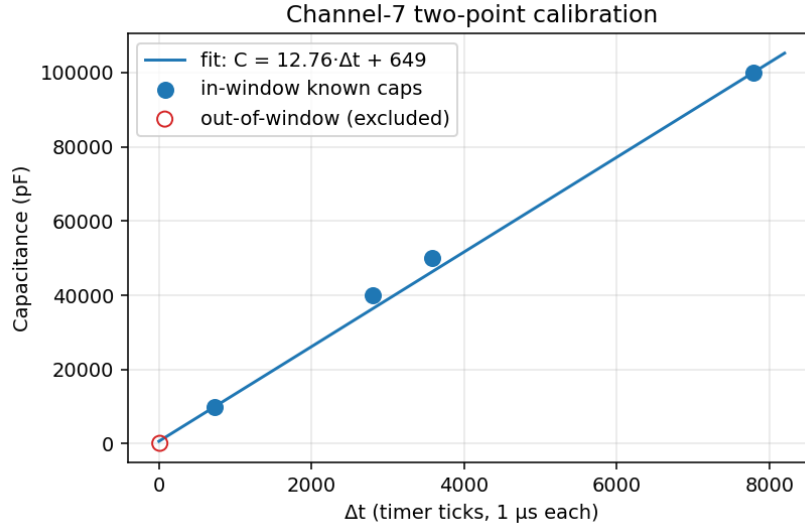


Figure 1: Channel-7 calibration. The in-window points (filled) define the affine fit; the out-of-window 100 pF point (open) illustrates why such points are excluded. The non-zero intercept is the offset b .

The figure was produced by the two-point fit below (run separately; the report embeds the resulting image so it renders without a Jupyter backend):

```
import numpy as np, matplotlib.pyplot as plt
data = [(733,10000),(3579,50000),(7788,100000),(2802,40000)] # in-window (Δt, C_pF)
a = (100000-10000)/(7788-733) # = 12.76 pF/tick
b = 10000 - a*733 # = 650 pF
# C = a·Δt + b (see eq. for the affine calibration model)
```

5.2 Raw measurement data and repeatability

Each value above is not a single shot but the **trimmed mean of 30 charge cycles** (the three highest and three lowest discarded, the middle 24 averaged); the median is reported alongside as a cross-check. Table 2 summarises the repeatability of the channel-7 runs.

Table 2: Channel-7 repeatability over 30 cycles. The close agreement of trimmed mean and median confirms a stable central estimate despite the per-shot scatter.

Capacitor	trimmed mean Δt	min	max	median	spread (max – min)/
100 pF	12	12	13	12	(out of window)
10 nF	733	696	760	736	8.7 %
“104” ceramic	2802	2665	3360	2802	24.8 %
50 nF	3579	2943	4048	3587	30.9 %
100 nF	7788	6780	9228	7777	31.4 %

The single-shot scatter (the comparator dithering at the slow crossing, §6.2) is visible in the complete channel-7 runs below (Δt in timer ticks, 30 cycles each):

```
100 pF (out of window - only ~12 ticks, kept for completeness):
 13 12 12 12 12 12 13 13 13 12 12 12 12 12
 12 12 12 12 12 12 12 12 13 12 13 13 13 12 12      -> 12

10 nF [calibration anchor]:
 732 699 709 735 746 749 744 731 759 738 721 707 719 729 732
 737 760 749 736 734 742 727 696 715 740 740 738 713 755 744  -> 733
```

```

104 ceramic (~40 nF; DC-bias loss):
  3360 2700 2881 2809 2826 2744 2774 2903 2799 2811 2828 2802 2734 2813 2882
  2777 3082 2790 2839 2755 2795 2698 2790 2665 2771 3088 2670 2851 2812 2774 -> 2802

50 nF:
  3676 3637 3590 3695 3651 3516 3661 3706 3344 4048 3572 3587 3550 3491 3501
  3495 3423 3508 3547 3510 3721 3511 3714 3601 3648 2943 3493 3360 3616 3758 -> 3579

100 nF [calibration anchor]:
  9228 7637 7661 7940 8097 7713 7864 7524 7536 8687 7775 7954 8048 7912 7807
  7972 7681 7687 7586 7674 7837 8230 7877 7866 7777 7759 6873 6780 7659 7594 -> 7788

```

The two **anchors** marked above — (733, 10 nF) and (7788, 100 nF) — are the points fed to the two-point calibration; trimming discards the obvious outliers (e.g. the 9228 and 6780 in the 100 nF run) before the fit. The 50 nF and 104 runs serve as independent checks, and the 100 nF run shows directly why an out-of-window count (12 ticks) is unusable.

5.3 Automatic-ranging across seven decades

With ranging enabled and the channel-7 calibration applied, capacitors from 100 pF to 470 μF were measured (Table 3). The instrument selected an appropriate channel in every case.

Table 3: Auto-ranging results. The selection logic is correct throughout; accuracy is excellent wherever the multiplexer resistance is negligible.

Capacitor	Channel	Reading	Comment
0.33 μF	R3 (980 Ω)	0.337 μF	accurate
1 μF	R5 (9.86 kΩ)	0.975 μF	accurate
2.2 μF	R5	2.07 μF	within electrolytic tolerance
10 μF	R4 (5.5 kΩ)	8.76 μF	within electrolytic tolerance
47 μF	R3	46.5 μF	accurate
220 μF	R1 (56 Ω)	715 μF	gross error — see §6.1
680 μF	R0 (10 Ω)	overflow	beyond range — see §6.1

6 Discussion and error analysis

6.1 Multiplexer on-resistance

The two failures at the smallest resistors share a single cause: the analog multiplexer’s channel **on-resistance** R_{on} , which adds in series with the selected ladder resistor. The true charging resistance is $R_{\text{eff}} = R_{\text{nom}} + R_{\text{on}}$, and since calibration used the nominal value, low- Ω ranges over-read.

We can extract R_{on} from the data. Treating the 220 μF capacitor as nominal and inverting Equation 5,

$$R_{\text{eff}} = \frac{\Delta t}{C \ln 2} = \frac{31617 \mu\text{s}}{(220 \mu\text{F})(0.693)} \approx 207 \Omega,$$

so on channel R1 ($R_{\text{nom}} = 56 \Omega$) the multiplexer contributes $R_{\text{on}} \approx 150 \Omega$ — entirely consistent with the CD74HC4067 datasheet. The resulting error scales as $(R_{\text{nom}} + R_{\text{on}})/R_{\text{nom}}$:

Channel	R_{nom}	R_{eff}	uncorrected error
R0	10 Ω	160 Ω	16×
R1	56 Ω	206 Ω	3.7×
R3	980 Ω	1130 Ω	+15%
R4	5.5 kΩ	5.65 kΩ	+2.7%
R7	99.3 kΩ	99.45 kΩ	+0.15%

This table explains everything observed: the gross 3.7× over-reading at 220 μF (R1), the impossibility of measuring 680 μF (R_{on} raises R_{eff} until the shortest available time constant still overflows the timer), and equally *why channel 7 calibrated so*

cleanly — there the 150 Ω contributes a negligible 0.15%. The remedy is to populate the resistor table with **measured effective resistances** (the directly measurable pin-to-node resistance with each channel enabled) and, for the largest capacitors, to select a coarser timer prescaler to extend the upper count limit.

6.2 Random scatter and dielectric effects

Individual cycles scatter by $\pm 15\%$ because the comparator, lacking hysteresis, dithers at the slow crossing; the reported **trimmed mean** of 30 cycles (dropping the three highest and lowest) and the median agree to a few percent, confirming a stable central estimate. Notably, the instrument is *honest* about the components: the “104” ceramic, nominally 100 nF, measured 40 nF — matching a bench multimeter — because Class II ceramics lose capacitance under DC bias, whereas the film and electrolytic parts read within tolerance. The meter reports the *actual* capacitance at the test conditions, not the nameplate value.

6.3 Calibration simulation

To generalize the calibration across all ranges, the measurement was modelled in software as

$$\Delta t = p R_{\text{eff}} (C + C_{\text{stray}}), \quad R_{\text{eff}} = R_{\text{nom}} + R_{\text{on}}, \quad (11)$$

with $p = \ln(\text{ratio})/t_{\text{tick}}$ absorbing the clock and threshold-ratio errors. This is precisely the firmware’s own relation $C = \Delta t K/R + b$ inverted, with $K = 1/p$ and $b = -C_{\text{stray}}$ — so the simulation is structurally identical to the instrument’s algorithm. Given a trial R_{on} the model is linear in $(p, p C_{\text{stray}})$, fitted by weighted least squares (film references weighted above $\pm 20\%$ electrolytics).

The fit recovers a multiplexer resistance $R_{\text{on}} \approx 75\text{--}150 \Omega$ (consistent with the direct estimate of §6.1), a slope correction of 1.08 (within the internal oscillator’s $\pm 10\%$), and a stray offset of ≈ 600 pF that independently reproduces the C7 two-point intercept — confirming the offset is a genuine, range-independent stray. Figure 2 shows the eleven channels tiling roughly 47 pF–900 μF with healthy overlap; Figure 3 shows the modelled versus measured intervals.

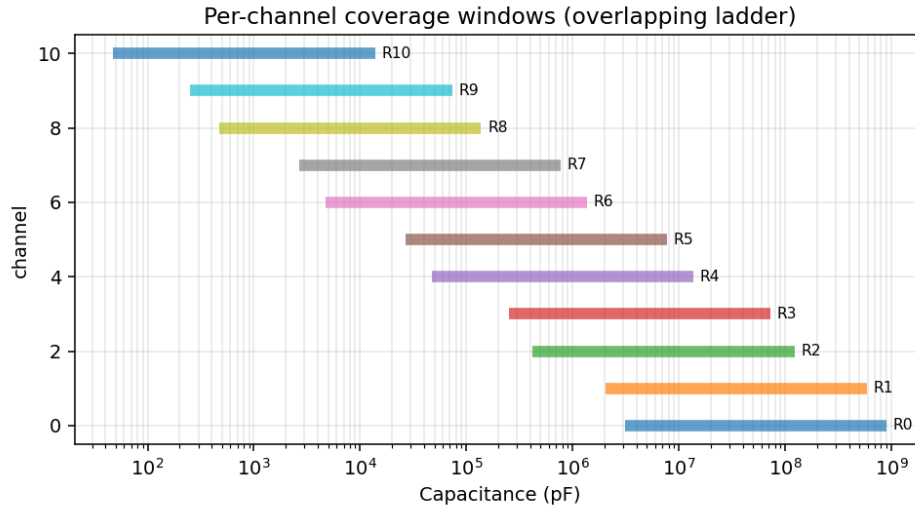


Figure 2: Per-channel coverage windows. Each resistor admits ~ 2.5 decades; the geometric ladder overlaps to span seven decades.

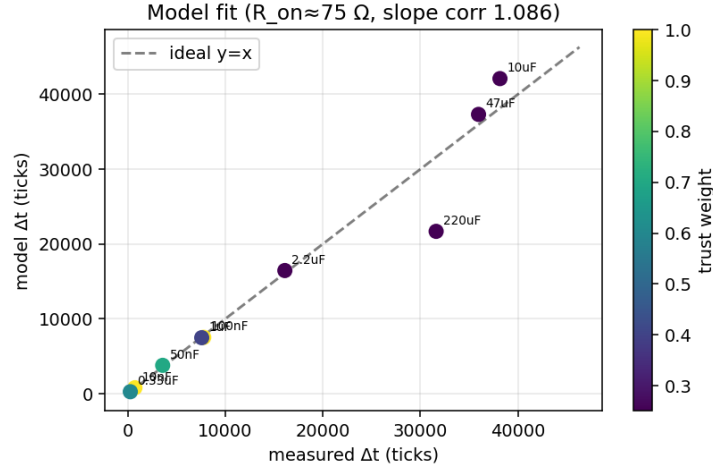


Figure 3: Modelled vs measured Δt , coloured by reference trust. Mid-range (film) points fit well; the low-trust electrolytic extremes ($\pm 20\%$) scatter — the limit on a precise cross-range fit.

The simulation’s value is therefore twofold: it validates the firmware equation and resistor ladder, and it isolates the two remaining error sources — multiplexer resistance and an under-determined offset. A definitive universal calibration awaits **film reference capacitors measured on the low-resistance channels**, which the $\pm 20\%$ electrolytics used here cannot provide.

7 Comparison with professional instruments

It is instructive to compare this DC charge-timing instrument with how laboratory-grade meters work, because the professional techniques map directly onto the error sources identified above.

AC auto-balancing bridge (LCR meters). Rather than a DC charge, an LCR meter applies a small *sinusoidal* test voltage at a chosen frequency and measures the resulting current with a transimpedance amplifier whose input is held at virtual ground. Since a capacitor’s impedance is $Z = 1/(j\omega C)$, the magnitude and phase of V/I — recovered by synchronous (lock-in) detection — yield C and the loss terms (ESR, dissipation factor D , quality factor Q) as the real and imaginary parts. This measures the part under its rated small-signal conditions and rejects noise far better than a single DC interval can.

Constant-current charge (bench DMMs). A simpler professional method charges the capacitor from a *constant current source* rather than through a resistor, so the voltage ramps **linearly**: $Q = It = CV \Rightarrow C = It/V$. The linear ramp avoids the logarithm of an exponential and removes the resistor-tolerance term altogether.

The decisive point is calibration. Each limitation we met has a standard professional remedy:

Our limitation	Professional remedy
Mux R_{on} / contact resistance (220 μF read $3\times$ high)	Kelvin (4-wire) sensing — separate force/sense leads carry no sense current through the series resistance; plus a SHORT calibration
Stray capacitance (600 pF offset)	OPEN calibration — measure with terminals open, subtract the stray
Clock error (internal RC $\pm 10\%$)	crystal / TCXO timebase (our planned 16 MHz upgrade)
Comparator jitter ($\pm 15\%$ scatter)	synchronous (lock-in) detection + averaging
Auto-ranging	switched precision feedback resistor <i>and</i> test frequency, calibrated out
Ceramic DC-bias loss (“104” $\rightarrow$ 40 nF)	programmable DC bias with C -vs-bias characterization

The headline is that **OPEN/SHORT calibration** — the cornerstone of every professional bridge — is precisely the formalization of our corrections: **OPEN** is our stray term C_{stray} , **SHORT** is our multiplexer R_{on} . In that sense our instrument occupies the *DC RC-timing* tier; the next step up is constant-current timing, and the professional tier is the AC auto-balancing bridge with Kelvin sensing, a crystal timebase, lock-in detection and OPEN/SHORT calibration — which additionally yields ESR and the behaviour versus frequency and bias that a single DC measurement cannot.

8 Conclusions and future work

We have realized a supply-independent, auto-ranging capacitance meter on a modest 8-bit microcontroller, validated from 100 pF to 470 μ F. The two-threshold timing method cleanly cancels the supply voltage (Equation 4), the bounded-counter and time-constant arguments together justify automatic ranging, and a two-point affine calibration (Equation 7) corrects the residual systematic errors. The dominant remaining limitation — multiplexer on-resistance at low ranges — has been identified and quantified at $R_{\text{on}} \approx 150 \, \Omega$.

Three refinements are proposed. First, replacing the nominal ladder values with **measured effective resistances** will recover accuracy on the low- Ω channels. Second, a **per-range timer prescaler** will extend the upper limit to the millifarad region. Third, a **multi-point least-squares calibration** across ranges — extending the simulation of §6.3 — will distinguish a constant stray-capacitance offset from a resistance-dependent timing offset and thereby deliver uniform accuracy across all seven decades.